

Topic B10

Diseases and immunity

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B10.1	Pathogens and transmission
B10.2	Body defences
B10.3	Immunity and vaccination
B10.4	Controlling disease

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Pathogens and transmission

A pathogen is a disease-causing organism and may be a bacterium, virus, fungus or protist. Transmission can be direct, by droplets or body fluids, or indirect through food, water, surfaces or vectors. Good sanitation, safe water, hygiene and vector control break transmission routes.

Examiner insight

Antibiotics act on bacteria, not viruses.

2. First-line defences

Skin forms a physical barrier and clotting seals wounds. Nasal hairs and mucus trap pathogens; cilia move mucus toward the throat. Stomach acid kills many swallowed microorganisms. These non-specific defences act before an immune response.

Examiner insight

Mucus traps pathogens; cilia move the mucus rather than killing pathogens directly.

3. Immune response

Phagocytes engulf and digest pathogens. Lymphocytes recognise antigens and produce specific antibodies; some form memory cells. On re-exposure, memory cells produce antibodies faster and in greater quantity, often preventing symptoms.

Examiner insight

Antibodies are specific because their binding sites complement particular antigens.

4. Vaccination

Vaccines contain harmless antigenic material that stimulates lymphocytes and memory-cell formation without causing the full disease. High coverage reduces transmission and indirectly protects susceptible people. Booster doses renew memory responses.

Examiner insight

Vaccination prevents disease by preparing the immune response; it does not directly kill every pathogen at injection.

Worked application

Situation: Vaccination coverage falls in a community.

Prediction: Outbreak risk rises.

Reason: Fewer immune people allow longer transmission chains and reduce indirect protection.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B10

Diseases and immunity

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines pathogen? [1]
 - A an organism that causes disease
 - B a molecule recognised as foreign that stimulates an immune response
 - C a specific protein produced by lymphocytes that binds an antigen
 - D the ability to resist a particular infection through an effective response
2. Which statement best defines antigen? [1]
 - A a specific protein produced by lymphocytes that binds an antigen
 - B a molecule recognised as foreign that stimulates an immune response
 - C the ability to resist a particular infection through an effective response
 - D introduction of harmless antigenic material to produce immunity
3. Which statement best defines antibody? [1]
 - A the ability to resist a particular infection through an effective response
 - B introduction of harmless antigenic material to produce immunity
 - C a specific protein produced by lymphocytes that binds an antigen
 - D an organism that causes disease
4. Which statement best defines immunity? [1]
 - A introduction of harmless antigenic material to produce immunity
 - B an organism that causes disease
 - C a molecule recognised as foreign that stimulates an immune response
 - D the ability to resist a particular infection through an effective response
5. Which statement best defines vaccination? [1]
 - A introduction of harmless antigenic material to produce immunity
 - B an organism that causes disease
 - C a molecule recognised as foreign that stimulates an immune response
 - D a specific protein produced by lymphocytes that binds an antigen
6. Which statement about pathogens and transmission is correct? [1]
 - A Any microorganism.
 - B A pathogen is a disease-causing organism and may be a bacterium, virus, fungus or protist.
 - C An antibody made by a pathogen.
 - D A chemical that kills all cells.
7. Which statement about first-line defences is correct? [1]
 - A An antibody made by a pathogen.
 - B A chemical that kills all cells.
 - C Skin forms a physical barrier and clotting seals wounds.
 - D Absence of all microorganisms.

8. Which statement about immune response is correct? [1]
- A A chemical that kills all cells.
 - B Absence of all microorganisms.
 - C Treatment of viral disease with antibiotics.
 - D Phagocytes engulf and digest pathogens.
9. Which statement about vaccination is correct? [1]
- A Vaccines contain harmless antigenic material that stimulates lymphocytes and memory-cell formation without causing the full disease.
 - B Absence of all microorganisms.
 - C Treatment of viral disease with antibiotics.
 - D Any microorganism.
10. Which prediction is most appropriate when vaccination coverage falls in a community? [1]
- A Any microorganism.
 - B A chemical that kills all cells.
 - C Outbreak risk rises.
 - D Treatment of viral disease with antibiotics.

Section B · Structured questions

11 Biologists use precise vocabulary when studying diseases and immunity.

(a) Define pathogen.

[2]

(b) Define antigen.

[2]

(c) Distinguish antibody from the incorrect idea: “a chemical that kills all cells”.

[2]

(d) State two important features of pathogens and transmission.

[2]

[Total: 8]

12 Vaccination coverage falls in a community.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative peak antibody level under three conditions.

condition	relative peak antibody level
first exposure	3
second exposure	14
third exposure	18

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to compare microbial growth near antiseptic discs.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of diseases and immunity to an unfamiliar context.

(a) Define immunity. [2]

(b) Explain the biological importance of immune response. [3]

(c) Correct this statement: "treatment of viral disease with antibiotics". [2]

(d) State one further question that could be investigated. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. An organism that causes disease.
- 2 B. A molecule recognised as foreign that stimulates an immune response.
- 3 C. A specific protein produced by lymphocytes that binds an antigen.
- 4 D. The ability to resist a particular infection through an effective response.
- 5 A. Introduction of harmless antigenic material to produce immunity.
- 6 B. Antibiotics act on bacteria, not viruses.
- 7 C. Mucus traps pathogens; cilia move the mucus rather than killing pathogens directly.
- 8 D. Antibodies are specific because their binding sites complement particular antigens.
- 9 A. Vaccination prevents disease by preparing the immune response; it does not directly kill every pathogen at injection.
- 10 C. Fewer immune people allow longer transmission chains and reduce indirect protection.

Solutions · Structured questions

11. (a) an organism that causes disease [2]. (b) a molecule recognised as foreign that stimulates an immune response [2]. (c) a specific protein produced by lymphocytes that binds an antigen; not a chemical that kills all cells [2]. (d) Credit any two accurate features from the guide: A pathogen is a disease-causing organism and may be a bacterium, virus, fungus or protoctist. Transmission can be direct, by droplets or body fluids, or indirect through food, water, surfaces or vectors. Good sanitation, safe water, hygiene and vector control break transmission routes. [2].

12. (a) Outbreak risk rises [2]. (b) Fewer immune people allow longer transmission chains and reduce indirect protection [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $18 - 3 = 15$ [2]. (b) The values change from 3 to 18, with the highest value 18. [2]. (c) Memory cells cause faster and larger secondary responses [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Antiseptic type [1]. (b) Diameter of inhibition zone [1]. (c) Same microorganism lawn, disc size, concentration, agar and incubation [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use aseptic technique, replicate plates and keep lids closed [1].

15. (a) the ability to resist a particular infection through an effective response [2]. (b) Award linked points from: Phagocytes engulf and digest pathogens. Lymphocytes recognise antigens and produce specific antibodies; some form memory cells. On re-exposure, memory cells produce antibodies faster and in greater quantity, often preventing symptoms. [3]. (c) Correct statement: vaccination means introduction of harmless antigenic material to produce immunity [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.